JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Groups (2005–2026 Archive)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
101	214 M	303 Min	MCQ • MSQ • NAT (2005–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	56	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	16	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	29	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q47 • ID: JAM_2026_Q47) NAT +1 M Neg: Nil (0)

Q.47	Let C_n denote a cyclic group having n elements. If there is a surjective group homomorphism from C_n to C_{30}, then the total number of such distinct surjective homomorphisms is ______.

Question 2 (JAM 2026 • Q30 • ID: JAM_2026_Q30) MCQ +2 M Neg: −0.67

Q.30	Let G be the power set of $\{1,2,3,4\}$. Let the binary operation Δ on G be $A \,\Delta\, B = (A \setminus B) \cup (B \setminus A)$. Which ONE of the following statements is TRUE?
(A)	$\{1\}$ is an identity element of (G, Δ).
(B)	(G, Δ) is an abelian group but not cyclic.
(C)	(G, Δ) is a group and has an element of order 4.
(D)	(G, Δ) is a group and has an element of order 8.

Question 3 (JAM 2026 • Q27 • ID: JAM_2026_Q27) **MCQ** **+2 M** **Neg: −0.67**

Q.27	Which ONE of the following statements is FALSE?
(A)	There is a surjective group homomorphism from the additive group of rational numbers to the multiplicative group of all complex roots of unity.
(B)	The multiplicative group of complex numbers of modulus one is isomorphic to a quotient group of the additive group of real numbers.
(C)	Any group homomorphism from the multiplicative group of nonzero complex numbers into the group of all invertible 2×2 matrices with real entries has nontrivial kernel.
(D)	There exists a group homomorphism from the symmetric group on n symbols into the multiplicative group of all invertible $n \times n$ matrices with real entries, which has a trivial kernel.

Question 4 (JAM 2026 • Q26 • ID: JAM_2026_Q26) MCQ +2 M Neg: −0.67

Q.26	Let $f: G \longrightarrow \mathbb{Z}$ be a surjective group homomorphism from an abelian group G to the additive group of integers $\mathbb{Z}$. Let K denote the kernel of f. Which ONE of the following statements is FALSE?
(A)	There exists a group homomorphism $\phi: \mathbb{Z} \longrightarrow G$ such that $f \circ \phi$ is the identity homomorphism.
(B)	K is isomorphic to a quotient group of G.
(C)	For all non-zero homomorphisms $\alpha: \mathbb{Z} \longrightarrow G$ and $\beta: G \longrightarrow K$, the composition $\beta \circ \alpha$ is non-zero.
(D)	G has a subgroup isomorphic to $\mathbb{Z}$.

Question 5 (JAM 2026 • Q9 • ID: JAM_2026_Q9) MCQ +1 M Neg: −0.33

Q.9	Which ONE of the following statements is FALSE?
(A)	S_3 is isomorphic to a subgroup of S_4.
(B)	$\mathbb{Z}_3$ is isomorphic to a subgroup of S_4.
(C)	S_3 is isomorphic to a quotient group of S_4.
(D)	$\mathbb{Z}_6$ is isomorphic to a quotient group of S_4.

Question 6 (JAM 2025 • Q60 • ID: JAM_2025_Q60) **NAT** **+2 M** **Neg: Nil (0)**

Q.60 The number of surjective (onto) group homomorphisms from S_4 to $\mathbb{Z}_6$ is equal to _______________.

Question 7 (JAM 2025 • Q59 • ID: JAM_2025_Q59) **NAT** **+2 M** **Neg: Nil (0)**

Q.59 Let G be an abelian group of order 35. Let m denote the number of elements of order 5 in G, and let n denote the number of elements of order 7 in G.

Then, the value of $m + n$ is equal to _______________.

Question 8 (JAM 2025 • Q52 • ID: JAM_2025_Q52) **NAT** **+2 M** **Neg: Nil (0)**

Q.52 Let $\sigma \in S_4$ be the permutation defined by $\sigma(1) = 2,\ \sigma(2) = 3,\ \sigma(3) = 1$ and $\sigma(4) = 4$. The number of elements in the set

$$\{\tau \in S_4 : \tau\sigma\tau^{-1} = \sigma\}$$

is equal to _______________.

Question 9 (JAM 2025 • Q39 • ID: JAM_2025_Q39) **MSQ** **+2 M** **Neg: Nil (0)**

Q.39 For $n \in \mathbb{N}$, consider the set $U(n) = \{\bar{x} \in \mathbb{Z}_n : \ \gcd(x, n) = 1\}$ as a group under multiplication modulo n .

Then, which of the following is/are TRUE?

(A) $U(8)$ is a cyclic group

(B) $U(5)$ is a cyclic group

(C) $U(12)$ is a cyclic group

(D) $U(9)$ is a cyclic group

Question 10 (JAM 2025 • Q23 • ID: JAM_2025_Q23) **MCQ** **+2 M** **Neg: −0.67**

Q.23 Let $\mathbb{R}/\mathbb{Z}$ denote the quotient group, where $\mathbb{Z}$ is considered as a subgroup of the additive group of real numbers $\mathbb{R}$.

Let m denote the number of injective (one-one) group homomorphisms from $\mathbb{Z}_3$ to $\mathbb{R}/\mathbb{Z}$ and n denote the number of group homomorphisms from $\mathbb{R}/\mathbb{Z}$ to $\mathbb{Z}_3$.

Then, which one of the following is TRUE?

(A) $m = 2$ and $n = 1$

(B) $m = 3$ and $n = 3$

(C) $m = 2$ and $n = 3$

(D) $m = 1$ and $n = 1$

Question 11 (JAM 2025 • Q9 • ID: JAM_2025_Q9) MCQ +1 M Neg: −0.33

Q.9 Let G be a finite abelian group of order 10. Let x_0 be an element of order 2 in G.

If $X = \{x \in G : x^3 = x_0\}$, then which one of the following is TRUE?

(A) X has exactly one element

(B) X has exactly two elements

(C) X has exactly three elements

(D) X is an empty set

Question 12 (JAM 2024 • Q40 • ID: JAM_2024_Q40) MSQ +2 M Neg: Nil (0)

Q.40 Consider

$$G = \{m + n\sqrt{2} : m, n \in \mathbb{Z}\}$$

as a subgroup of the additive group $\mathbb{R}$.

Which of the following statements is/are TRUE?

(A) G is a cyclic subgroup of $\mathbb{R}$ under addition

(B) $G \cap I$ is non-empty for every non-empty open interval $I \subseteq \mathbb{R}$

(C) G is a closed subset of $\mathbb{R}$

(D) G is isomorphic to the group $\mathbb{Z} \times \mathbb{Z}$, where the group operation in $\mathbb{Z} \times \mathbb{Z}$ is defined by $(m_1, n_1) + (m_2, n_2) = (m_1 + m_2, n_1 + n_2)$

Question 13 (JAM 2024 • Q36 • ID: JAM_2024_Q36) MSQ +2 M Neg: Nil (0)

Q.36 The center $Z(G)$ of a group G is defined as

$$Z(G) = \{x \in G : xg = gx \text{ for all } g \in G\}.$$

Let $|G|$ denote the order of G. Then, which of the following statements is/are TRUE for any group G?

(A) If G is non-abelian and $Z(G)$ contains more than one element, then the center of the quotient group $G/Z(G)$ contains only one element

(B) If $|G| \geq 2$, then there exists a non-trivial homomorphism from $\mathbb{Z}$ to G

(C) If $|G| \geq 2$ and G is non-abelian, then there exists a non-identity isomorphism from G to itself

(D) If $|G| = p^3$, where p is a prime number, then G is necessarily abelian

Question 14 (JAM 2024 • Q33 • ID: JAM_2024_Q33) MSQ +2 M Neg: Nil (0)

Q.33 Which of the following statements is/are TRUE?

(A) The additive group of real numbers is isomorphic to the multiplicative group of positive real numbers

(B) The multiplicative group of nonzero real numbers is isomorphic to the multiplicative group of nonzero complex numbers

(C) The additive group of real numbers is isomorphic to the multiplicative group of nonzero complex numbers

(D) The additive group of real numbers is isomorphic to the additive group of rational numbers

Question 15 (JAM 2024 • Q12 • ID: JAM_2024_Q12) MCQ +2 M Neg: −0.67

Q.12 Consider the group $G = \{A \in M_2(\mathbb{R}) : AA^T = I_2\}$ with respect to matrix multiplication. Let

$$Z(G) = \{A \in G : AB = BA, \text{ for all } B \in G\}.$$

Then, the cardinality of $Z(G)$ is

(A) 1

(B) 2

(C) 4

(D) Infinite

Question 16 (JAM 2024 • Q11 • ID: JAM_2024_Q11) MCQ +2 M Neg: −0.67

Q.11 Which one of the following groups has elements of order $1, 2, 3, 4, 5$ but does not have an element of order greater than or equal to 6 ?

(A) The alternating group A_6

(B) The alternating group A_5

(C) S_6

(D) S_5

Question 17 (JAM 2024 • Q8 • ID: JAM_2024_Q8) MCQ +1 M Neg: −0.33

Q.8 Let G be a finite group containing a non-identity element which is conjugate to its inverse. Then, which one of the following is TRUE?

(A) The order of G is necessarily even

(B) The order of G is not necessarily even

(C) G is necessarily cyclic

(D) G is necessarily abelian but need not be cyclic

Question 18 (JAM 2024 • Q7 • ID: JAM_2024_Q7) MCQ +1 M Neg: −0.33

Q.7 Which one of the following is TRUE for the symmetric group S_{13}?

(A) S_{13} has an element of order 42

(B) S_{13} has no element of order 35

(C) S_{13} has an element of order 27

(D) S_{13} has no element of order 60

Question 19 (JAM 2024 • Q6 • ID: JAM_2024_Q6) MCQ +1 M Neg: −0.33

Q.6 For a positive integer n, let $U(n) = \{\bar{r} \in \mathbb{Z}_n : \gcd(r,n) = 1\}$ be the group under multiplication modulo n. Then, which one of the following statements is TRUE?

(A) $U(5)$ is isomorphic to $U(8)$

(B) $U(10)$ is isomorphic to $U(12)$

(C) $U(8)$ is isomorphic to $U(10)$

(D) $U(8)$ is isomorphic to $U(12)$

Question 20 (JAM 2024 • Q5 • ID: JAM_2024_Q5) MCQ +1 M Neg: −0.33

Q.5 Let G be a group of order 39 such that it has exactly one subgroup of order 3 and exactly one subgroup of order 13. Then, which one of the following statements is TRUE?

(A) G is necessarily cyclic

(B) G is abelian but need not be cyclic

(C) G need not be abelian

(D) G has 13 elements of order 13

Question 21 (JAM 2023 • Q54 • ID: JAM_2023_Q54) NAT +2 M Neg: Nil (0)

Q. 54 The number of permutations in S_4 that have exactly two cycles in their cycle decompositions is equal to ____________.

Question 22 (JAM 2023 • Q46 • ID: JAM_2023_Q46) NAT +1 M Neg: Nil (0)

Q. 46 For $g \in \mathbb{Z}$, let $\bar{g} \in \mathbb{Z}_8$ denote the residue class of g modulo 8. Consider the group $\mathbb{Z}_8^\times = \{\bar{x} \in \mathbb{Z}_8 : 1 \leq x \leq 7, \gcd(x, 8) = 1\}$ with respect to multiplication modulo 8. The number of group isomorphisms from $\mathbb{Z}_8^\times$ onto itself is equal to ____________.

Question 23 (JAM 2023 • Q45 • ID: JAM_2023_Q45) NAT +1 M Neg: Nil (0)

Q. 45 For $\sigma \in S_8$, let $o(\sigma)$ denote the order of σ. Then $\max\{o(\sigma) : \sigma \in S_8\}$ is equal to ____________.

Question 24 (JAM 2023 • Q32 • ID: JAM_2023_Q32) **MSQ** **+2 M** **Neg: Nil (0)**

Q. 32 Let $A \subseteq \mathbb{Z}$ with $0 \in A$. For $r, s \in \mathbb{Z}$, define

$$rA = \{ra : a \in A\}, \qquad rA + sA = \{ra + sb : \ a, b \in A\}.$$

Which of the following conditions imply that A is a subgroup of the additive group $\mathbb{Z}$?

(A) $-2A \subseteq A, \ \ A + A = A$

(B) $A = -A, \ \ A + 2A = A$

(C) $A = -A, \ \ A + A = A$

(D) $2A \subseteq A, \ \ A + A = A$

Question 25 (JAM 2023 • Q29 • ID: JAM_2023_Q29) **MCQ** **+2 M** **Neg: −0.67**

Q. 29 From the additive group $\mathbb{Q}$ to which one of the following groups does there exist a non-trivial group homomorphism?

(A) $\mathbb{R}^{\times}$, the multiplicative group of non-zero real numbers

(B) $\mathbb{Z}$, the additive group of integers

(C) $\mathbb{Z}_2$, the additive group of integers modulo 2

(D) $\mathbb{Q}^{\times}$, the multiplicative group of non-zero rational numbers

Question 26 (JAM 2023 • Q27 • ID: JAM_2023_Q27) MCQ +2 M Neg: −0.67

Q. 27 How many group homomorphisms are there from $\mathbb{Z}_2$ to S_5?

(A) 40

(B) 41

(C) 26

(D) 25

Question 27 (JAM 2023 • Q23 • ID: JAM_2023_Q23) MCQ +2 M Neg: −0.67

Q. 23 Consider the following statements:

I. Every infinite group has infinitely many subgroups.

II. There are only finitely many non-isomorphic groups of a given finite order.

Then

(A) both I and II are TRUE

(B) I is TRUE but II is FALSE

(C) I is FALSE but II is TRUE

(D) both I and II are FALSE

Question 28 (JAM 2023 • Q1 • ID: JAM_2023_Q1) MCQ +1 M Neg: −0.33

Q. 1 Let G be a finite group. Then G is necessarily a cyclic group if the order of G is

(A) 4

(B) 7

(C) 6

(D) 10

Question 29 (JAM 2022 • Q58 • ID: JAM_2022_Q58) NAT +2 M Neg: Nil (0)

Q.58 Let σ be the permutation in the symmetric group S_5 given by

$$\sigma(1) = 2, \quad \sigma(2) = 3, \quad \sigma(3) = 1, \quad \sigma(4) = 5, \quad \sigma(5) = 4.$$

Define

$$N(\sigma) = \{\tau \in S_5 : \sigma \circ \tau = \tau \circ \sigma\}.$$

Then the number of elements in $N(\sigma)$ is equal to ________.

Question 30 (JAM 2022 • Q46 • ID: JAM_2022_Q46) NAT +1 M Neg: Nil (0)

Q.46 The number of elements of order 12 in the symmetric group S_7 is equal to ________.

Question 31 (JAM 2022 • Q45 • ID: JAM_2022_Q45) NAT +1 M Neg: Nil (0)

Q.45 The number of distinct subgroups of $\mathbb{Z}_{999}$ is ________.

Question 32 (JAM 2022 • Q37 • ID: JAM_2022_Q37) MSQ +2 M Neg: Nil (0)

Q.37	Let G be a finite group of order at least two and let e denote the identity element of G. Let $\sigma: G \to G$ be a bijective group homomorphism that satisfies the following two conditions: (i) If $\sigma(g) = g$ for some $g \in G$, then $g = e$, (ii) $(\sigma \circ \sigma)(g) = g$ for all $g \in G$. Then which of the following is/are correct?
(A)	For each $g \in G$, there exists $h \in G$ such that $h^{-1}\sigma(h) = g$.
(B)	There exists $x \in G$ such that $x\sigma(x) \neq e$.
(C)	The map σ satisfies $\sigma(x) = x^{-1}$ for every $x \in G$.
(D)	The order of the group G is an odd number.

Question 33 (JAM 2022 • Q24 • ID: JAM_2022_Q24) MCQ +2 M Neg: −0.67

Q.24	For $g \in \mathbb{Z}$, let $\bar{g} \in \mathbb{Z}_{37}$ denote the residue class of g modulo 37. Consider the group $U_{37} = \{\bar{g} \in \mathbb{Z}_{37} : 1 \le g \le 37 \text{ with } \gcd(g, 37) = 1\}$ with respect to multiplication modulo 37. Then which one of the following is FALSE?
(A)	The set $\{\bar{g} \in U_{37} : \bar{g} = (\bar{g})^{-1}\}$ contains exactly 2 elements.
(B)	The order of the element $\overline{10}$ in U_{37} is 36.
(C)	There is exactly one group homomorphism from U_{37} to $(\mathbb{Z}, +)$.
(D)	There is exactly one group homomorphism from U_{37} to $(\mathbb{Q}, +)$.

Question 34 (JAM 2022 • Q19 • ID: JAM_2022_Q19) MCQ +2 M Neg: −0.67

Q.19	Consider the group $(\mathbb{Q}, +)$ and its subgroup $(\mathbb{Z}, +)$. For the quotient group $\mathbb{Q}/\mathbb{Z}$, which one of the following is FALSE?
(A)	$\mathbb{Q}/\mathbb{Z}$ contains a subgroup isomorphic to $(\mathbb{Z}, +)$.
(B)	There is exactly one group homomorphism from $\mathbb{Q}/\mathbb{Z}$ to $(\mathbb{Q}, +)$.
(C)	For all $n \in \mathbb{N}$, there exists $g \in \mathbb{Q}/\mathbb{Z}$ such that the order of g is n.
(D)	$\mathbb{Q}/\mathbb{Z}$ is not a cyclic group.

Question 35 (JAM 2022 • Q4 • ID: JAM_2022_Q4) MCQ +1 M Neg: −0.33

Q.4	Let G be a group of order 2022. Let H and K be subgroups of G of order 337 and 674, respectively. If $H \cup K$ is also a subgroup of G, then which one of the following is FALSE?
(A)	H is a normal subgroup of $H \cup K$.
(B)	The order of $H \cup K$ is 1011.
(C)	The order of $H \cup K$ is 674.
(D)	K is a normal subgroup of $H \cup K$.

Question 36 (JAM 2021 • Q51 • ID: JAM_2021_Q51) NAT +2 M Neg: Nil (0)

Q. 51 The number of elements of order two in the group S_4 is equal to _____.

Question 37 (JAM 2021 • Q47 • ID: JAM_2021_Q47) NAT +1 M Neg: Nil (0)

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Q. 47 The number of group homomorphisms from the group $\mathbb{Z}_4$ to the group S_3 is _____.

Question 38 (JAM 2021 • Q41 • ID: JAM_2021_Q41) NAT +1 M Neg: Nil (0)

Q. 41 The number of cycles of length 4 in S_6 is _____.

Question 39 (JAM 2021 • Q36 • ID: JAM_2021_Q36) MSQ +2 M Neg: Nil (0)

Q. 36 Let G be a finite group of order 28. Assume that G contains a subgroup of order 7. Which of the following statements is/are true?

(A) G contains a unique subgroup of order 7.

(B) G contains a normal subgroup of order 7.

(C) G contains no normal subgroup of order 7.

(D) G contains at least two subgroups of order 7.

Question 40 (JAM 2021 • Q28 • ID: JAM_2021_Q28) MCQ +2 M Neg: −0.67

Q. 28 Let G be a finite abelian group of odd order. Consider the following two statements:

I. The map $f : G \to G$ defined by $f(g) = g^2$ is a group isomorphism.

II. The product $\prod_{g \in G} g = e$.

(A) Both I and II are TRUE.

(B) I is TRUE but II is FALSE.

(C) II is TRUE but I is FALSE.

(D) Neither I nor II is TRUE.

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Question 41 (JAM 2021 • Q27 • ID: JAM_2021_Q27) MCQ +2 M Neg: −0.67

Q. 27 Let g be an element of S_7 such that g commutes with the element $(2, 6, 4, 3)$. The number of such g is

(A) 6. (B) 4. (C) 24. (D) 48.

Question 42 (JAM 2021 • Q24 • ID: JAM_2021_Q24) MCQ +2 M Neg: −0.67

Q. 24 Which one of the following statements is true?

(A) $(\mathbb{Z}, +)$ is isomorphic to $(\mathbb{R}, +)$.

(B) $(\mathbb{Z}, +)$ is isomorphic to $(\mathbb{Q}, +)$.

(C) $(\mathbb{Q}/\mathbb{Z}, +)$ is isomorphic to $(\mathbb{Q}/2\mathbb{Z}, +)$.

(D) $(\mathbb{Q}/\mathbb{Z}, +)$ is isomorphic to $(\mathbb{Q}, +)$.

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Question 43 (JAM 2021 • Q20 • ID: JAM_2021_Q20) MCQ +2 M Neg: −0.67

Q. 20 Which one of the following statements is true?

(A) Exactly half of the elements in any even order subgroup of S_5 must be even permutations.

(B) Any abelian subgroup of S_5 is trivial.

(C) There exists a cyclic subgroup of S_5 of order 6.

(D) There exists a normal subgroup of S_5 of index 7.

Question 44 (JAM 2021 • Q14 • ID: JAM_2021_Q14) MCQ +2 M Neg: −0.67

Q. 14 Consider the following statements.

I. The group $(\mathbb{Q}, +)$ has no proper subgroup of finite index.

II. The group $(\mathbb{C} \setminus \{0\}, \cdot)$ has no proper subgroup of finite index.

Which one of the following statements is true?

(A) Both I and II are TRUE.

(B) I is TRUE but II is FALSE.

(C) II is TRUE but I is FALSE.

(D) Neither I nor II is TRUE.

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Question 45 (JAM 2021 • Q6 • ID: JAM_2021_Q6) MCQ +1 M Neg: −0.33

Q. 6 How many elements of the group $\mathbb{Z}_{50}$ have order 10?

(A) 10 (B) 4 (C) 5 (D) 8

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Question 46 (JAM 2020 • Q60 • ID: JAM_2020_Q60) **NAT** **+2 M** **Neg: Nil (0)**

Q. 60 Suppose that G is a group of order 57 which is NOT cyclic. If G contains a unique subgroup H of order 19, then for any $g \notin H$, $o(g)$ is ________.

END OF THE QUESTION PAPER

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Question 47 (JAM 2020 • Q50 • ID: JAM_2020_Q50) **NAT** **+1 M** **Neg: Nil (0)**

Q. 50 Let $\phi : S_3 \to S^1$ be a non-trivial non-injective group homomorphism. Then, the number of elements in the kernel of ϕ is ________.

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Question 48 (JAM 2020 • Q40 • ID: JAM_2020_Q40) **MSQ** **+2 M** **Neg: Nil (0)**

Q. 40 Let G be a group with identity e. Let H be an abelian non-trivial proper subgroup of G with the property that $H \cap gHg^{-1} = \{e\}$ for all $g \notin H$.

If $K = \{g \in G : gh = hg \text{ for all } h \in H\}$, then

(A) K is a proper subgroup of H

(B) H is a proper subgroup of K

(C) $K = H$

(D) there exists no abelian subgroup $L \subseteq G$ such that K is a proper subgroup of L

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Question 49 (JAM 2020 • Q30 • ID: JAM_2020_Q30) MCQ +2 M Neg: −0.67

Q. 30 Let $F = \{\omega \in \mathbb{C} : \omega^{2020} = 1\}$. Consider the groups

$$G = \left\{ \begin{pmatrix} \omega & z \\ 0 & 1 \end{pmatrix} : \omega \in F, z \in \mathbb{C} \right\}$$

and

$$H = \left\{ \begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix} : z \in \mathbb{C} \right\}$$

under matrix multiplication. Then the number of cosets of H in G is

(A) 1010 (B) 2019 (C) 2020 (D) infinite

Question 50 (JAM 2020 • Q29 • ID: JAM_2020_Q29) **MCQ** **+2 M** **Neg: −0.67**

Q. 29 Let $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ be the circle group under multiplication and $i = \sqrt{-1}$. Then the set $\{\theta \in \mathbb{R} : \langle e^{i2\pi\theta} \rangle \text{ is infinite}\}$ is

(A) empty

(B) non-empty and finite

(C) countably infinite

(D) uncountable

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Question 51 (JAM 2020 • Q10 • ID: JAM_2020_Q10) **MCQ** **+1 M** **Neg: −0.33**

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Q. 10 Consider the following group under matrix multiplication:

$$H = \left\{ \begin{bmatrix} 1 & p & q \\ 0 & 1 & r \\ 0 & 0 & 1 \end{bmatrix} : p, q, r \in \mathbb{R} \right\}.$$

Then the center of the group is isomorphic to

(A) $(\mathbb{R} \setminus \{0\}, \times)$

(B) $(\mathbb{R}, +)$

(C) $(\mathbb{R}^2, +)$

(D) $(\mathbb{R}, +) \times (\mathbb{R} \setminus \{0\}, \times)$

Q. 11 – Q. 30 carry two marks each.

Question 52 (JAM 2019 • Q55 • ID: JAM_2019_Q55) **NAT** **+2 M** **Neg: Nil (0)**

Q.55 The number of elements in the set $\{x \in S_3 : x^4 = e\}$, where e is the identity element of the permutation group S_3, is ___________

Question 53 (JAM 2019 • Q53 • ID: JAM_2019_Q53) NAT +2 M Neg: Nil (0)

Q.53 Let $G = \{n \in \mathbb{N} : n \leq 55, \gcd(n, 55) = 1\}$ be the group under multiplication modulo 55. Let $x \in G$ be such that $x^2 = 26$ and $x > 30$. Then x is equal to____________

Question 54 (JAM 2019 • Q41 • ID: JAM_2019_Q41) NAT +1 M Neg: Nil (0)

Q. 41 – Q. 50 carry one mark each.

Q.41 Let x be the 100-cycle $(1\ \ 2\ \ 3\ \cdots\ 100)$ and let y be the transposition $(49\ \ 50)$ in the permutation group S_{100}. Then the order of xy is ______

Question 55 (JAM 2019 • Q32 • ID: JAM_2019_Q32) MSQ +2 M Neg: Nil (0)

Q.32 Let G be a nonabelian group, $y \in G$, and let the maps f, g, h from G to itself be defined by

$$f(x) = yxy^{-1},\ g(x) = x^{-1} \text{ and } h = g \circ g.$$

Then

(A) g and h are homomorphisms and f is not a homomorphism
(B) h is a homomorphism and g is not a homomorphism
(C) f is a homomorphism and g is not a homomorphism
(D) f, g and h are homomorphisms

Question 56 (JAM 2019 • Q31 • ID: JAM_2019_Q31) MSQ +2 M Neg: Nil (0)

Q. 31 – Q. 40 carry two marks each.

Q.31 Let G be a noncyclic group of order 4. Consider the statements I and II:

I. There is NO injective (one-one) homomorphism from G to $\mathbb{Z}_8$
II. There is NO surjective (onto) homomorphism from $\mathbb{Z}_8$ to G

Then

(A) I is true
(B) I is false
(C) II is true
(D) II is false

Question 57 (JAM 2019 • Q11 • ID: JAM_2019_Q11) MCQ +2 M Neg: −0.67

Q.11 Let H and K be subgroups of $\mathbb{Z}_{144}$. If the order of H is 24 and the order of K is 36, then the order of the subgroup $H \cap K$ is

(A) 3 (B) 4 (C) 6 (D) 12

Question 58 (JAM 2018 • Q48 • ID: JAM_2018_Q48) NAT +1 M Neg: Nil (0)

Q.48 Let A_6 be the group of even permutations of 6 distinct symbols. Then the number of elements of order 6 in A_6 is ________________

Question 59 (JAM 2018 • Q41 • ID: JAM_2018_Q41) NAT +1 M Neg: Nil (0)

Q. 41 – Q. 50 carry one mark each.

Q.41 The order of the element $(1\ 2\ 3)\,(2\ 4\ 5)\,(4\ 5\ 6)$ in the group S_6 is ________________

Question 60 (JAM 2018 • Q35 • ID: JAM_2018_Q35) MSQ +2 M Neg: Nil (0)

Q.35 Let $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ denote the group of non-zero complex numbers under multiplication. Suppose $Y_n = \{ z \in \mathbb{C} \mid z^n = 1\},\ n \in \mathbb{N}$. Which of the following is (are) subgroup(s) of $\mathbb{C}^*$?

(A) $\bigcup_{n=1}^{100} Y_n$ (B) $\bigcup_{n=1}^{\infty} Y_{2^n}$ (C) $\bigcup_{n=100}^{\infty} Y_n$ (D) $\bigcup_{n=1}^{\infty} Y_n$

Question 61 (JAM 2018 • Q33 • ID: JAM_2018_Q33) MSQ +2 M Neg: Nil (0)

Q.33 Suppose f, g, h are permutations of the set $\{\alpha, \beta, \gamma, \delta\}$, where

f interchanges α and β but fixes γ and δ,

g interchanges β and γ but fixes α and δ,

h interchanges γ and δ but fixes α and β.

Which of the following permutations interchange(s) α and δ but fix(es) β and γ?

(A) $f \circ g \circ h \circ g \circ f$ (B) $g \circ h \circ f \circ h \circ g$ (C) $g \circ f \circ h \circ f \circ g$ (D) $h \circ g \circ f \circ g \circ h$

Question 62 (JAM 2018 • Q28 • ID: JAM_2018_Q28) MCQ +2 M Neg: −0.67

Q.28 Consider the group $\mathbb{Z}^2 = \{(a,b) \mid a, b \in \mathbb{Z}\}$ under component-wise addition. Then which of the following is a subgroup of $\mathbb{Z}^2$?

(A) $\{(a,b) \in \mathbb{Z}^2 \mid ab = 0\}$

(B) $\{(a,b) \in \mathbb{Z}^2 \mid 3a + 2b = 15\}$

(C) $\{(a,b) \in \mathbb{Z}^2 \mid 7 \text{ divides } ab\}$

(D) $\{(a,b) \in \mathbb{Z}^2 \mid 2 \text{ divides } a \text{ and } 3 \text{ divides } b\}$

Question 63 (JAM 2018 • Q26 • ID: JAM_2018_Q26) MCQ +2 M Neg: −0.67

Q.26 Let H be the quotient group $\mathbb{Q}/\mathbb{Z}$. Consider the following statements.

I. Every cyclic subgroup of H is finite.

II. Every finite cyclic group is isomorphic to a subgroup of H.

Which one of the following holds?

(A) I is TRUE but II is FALSE (B) II is TRUE but I is FALSE

(C) both I and II are TRUE (D) neither I nor II is TRUE

Question 64 (JAM 2018 • Q25 • ID: JAM_2018_Q25) MCQ +2 M Neg: −0.67

Q.25 Let G be a group satisfying the property that $f: G \to \mathbb{Z}_{221}$ is a homomorphism implies $f(g) = 0,\ \forall g \in G$. Then a possible group G is

(A) $\mathbb{Z}_{21}$ (B) $\mathbb{Z}_{51}$ (C) $\mathbb{Z}_{91}$ (D) $\mathbb{Z}_{119}$

MA 6/12

Question 65 (JAM 2018 • Q1 • ID: JAM_2018_Q1) MCQ +1 M Neg: −0.33

Q. 1 – Q.10 carry one mark each.

Q.1 Which one of the following is TRUE?

(A) $\mathbb{Z}_n$ is cyclic if and only if n is prime

(B) Every proper subgroup of $\mathbb{Z}_n$ is cyclic

(C) Every proper subgroup of S_4 is cyclic

(D) If every proper subgroup of a group is cyclic, then the group is cyclic

Question 66 (JAM 2017 • Q54 • ID: JAM_2017_Q54) NAT +2 M Neg: Nil (0)

Q.54 The maximum order of a permutation σ in the symmetric group S_{10} is ________

Question 67 (JAM 2017 • Q50 • ID: JAM_2017_Q50) NAT +1 M Neg: Nil (0)

Q.50 The number of subgroups of $\mathbb{Z}_7 \times \mathbb{Z}_7$ of order 7 is ________

Question 68 (JAM 2017 • Q43 • ID: JAM_2017_Q43) NAT +1 M Neg: Nil (0)

Q.43 Consider the permutations $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 5 & 3 & 7 & 8 & 6 & 1 & 2 \end{pmatrix}$ and $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 5 & 3 & 1 & 7 & 6 & 8 & 2 \end{pmatrix}$ in S_8. The number of $\eta \in S_8$ such that $\eta^{-1}\sigma\eta = \tau$ is equal to ________

Question 69 (JAM 2017 • Q42 • ID: JAM_2017_Q42) NAT +1 M Neg: Nil (0)

Q.42 Let G be a subgroup of $GL_2(\mathbb{R})$ generated by $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$. Then the order of G is

____

Question 70 (JAM 2017 • Q35 • ID: JAM_2017_Q35) MSQ +2 M Neg: Nil (0)

Q.35 Let G be a group of order 20 in which the conjugacy classes have sizes $1, 4, 5, 5, 5$. Then which of the following is/are TRUE?

(A) G contains a normal subgroup of order 5

(B) G contains a non-normal subgroup of order 5

(C) G contains a subgroup of order 10

(D) G contains a normal subgroup of order 4

Question 71 (JAM 2017 • Q31 • ID: JAM_2017_Q31) MSQ +2 M Neg: Nil (0)

Q.31 For $\alpha, \beta \in \mathbb{R}$, define the map $\varphi_{\alpha,\beta}: \mathbb{R} \to \mathbb{R}$ by $\varphi_{\alpha,\beta}(x) = \alpha x + \beta$. Let

$$G = \{\varphi_{\alpha,\beta} \mid (\alpha, \beta) \in \mathbb{R}^2\}$$

For $f, g \in G$, define $g \circ f \in G$ by $(g \circ f)(x) = g(f(x))$. Then which of the following statements is/are TRUE?

(A) The binary operation $\circ$ is associative

(B) The binary operation $\circ$ is commutative

(C) For every $(\alpha, \beta) \in \mathbb{R}^2$, $\alpha \neq 0$ there exists $(a, b) \in \mathbb{R}^2$ such that $\varphi_{\alpha,\beta} \circ \varphi_{a,b} = \varphi_{1,0}$

(D) $(G, \circ)$ is a group

Question 72 (JAM 2017 • Q3 • ID: JAM_2017_Q3) MCQ +1 M Neg: −0.33

Q.3 The number of generators of the additive group $\mathbb{Z}_{36}$ is equal to

(A) 6 (B) 12 (C) 18 (D) 36

Question 73 (JAM 2016 • Q58 • ID: JAM_2016_Q58) NAT +2 M Neg: Nil (0)

Q.58 Let H denote the group of all 2×2 invertible matrices over $\mathbb{Z}_5$ under usual matrix multiplication. Then the order of the matrix $\begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ in H is ____________

Question 74 (JAM 2016 • Q50 • ID: JAM_2016_Q50) NAT +1 M Neg: Nil (0)

Q.50 Let G be a cyclic group of order 12. Then the number of non-isomorphic subgroups of G is ____________

Q. 51 – Q. 60 carry two marks each.

Question 75 (JAM 2016 • Q38 • ID: JAM_2016_Q38) MSQ +2 M Neg: Nil (0)

Q.38 Let G be a finite group and $o(G)$ denotes its order. Then which of the following statement(s) is(are) TRUE?

(A) G is abelian if $o(G) = pq$ where p and q are distinct primes
(B) G is abelian if every non identity element of G is of order 2
(C) G is abelian if the quotient group $\frac{G}{Z(G)}$ is cyclic, where $Z(G)$ is the center of G
(D) G is abelian if $o(G) = p^3$, where p is prime

Question 76 (JAM 2016 • Q29 • ID: JAM_2016_Q29) MCQ +2 M Neg: −0.67

Q.29 In the permutation group S_n $(n \geq 5)$, if H is the smallest subgroup containing all the 3-cycles, then which one of the following is TRUE?

(A) Order of H is 2
(B) Index of H in S_n is 2
(C) H is abelian
(D) $H = S_n$

MA 7/13

Question 77 (JAM 2016 • Q28 • ID: JAM_2016_Q28) MCQ +2 M Neg: −0.67

Q.28 The number of group homomorphisms from the cyclic group $\mathbb{Z}_4$ to the cyclic group $\mathbb{Z}_7$ is

(A) 7 (B) 3 (C) 2 (D) 1

Question 78 (JAM 2016 • Q4 • ID: JAM_2016_Q4) MCQ +1 M Neg: −0.33

Q.4 Let σ be an element of the permutation group S_5. Then the maximum possible order of σ is

(A) 5 (B) 6 (C) 10 (D) 15

Question 79 (JAM 2015 • Q52 • ID: JAM_2015_Q52) NAT +2 M Neg: Nil (0)

Q.12 Suppose G is a cyclic group and $\sigma, \tau \in G$ are such that order$(\sigma) = 12$ and order$(\tau) = 21$. Then the order of the smallest group containing σ and τ is _______________

Question 80 (JAM 2015 • Q47 • ID: JAM_2015_Q47) NAT +1 M Neg: Nil (0)

Q.7

If $5^{2015} \equiv n \pmod{11}$ and $n \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, then n is equal to __________

MA 14/16

Question 81 (JAM 2015 • Q43 • ID: JAM_2015_Q43) NAT +1 M Neg: Nil (0)

Q.3

The number of distinct normal subgroups of S_3 is ______________

Question 82 (JAM 2015 • Q32 • ID: JAM_2015_Q32) MSQ +2 M Neg: Nil (0)

Q.2

Which of the following statements is (are) true?

(A) $\mathbb{Z}_2 \oplus \mathbb{Z}_3$ is isomorphic to $\mathbb{Z}_6$

(B) $\mathbb{Z}_3 \oplus \mathbb{Z}_3$ is isomorphic to $\mathbb{Z}_9$

(C) $\mathbb{Z}_4 \oplus \mathbb{Z}_6$ is isomorphic to $\mathbb{Z}_{24}$

(D) $\mathbb{Z}_2 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_5$ is isomorphic to $\mathbb{Z}_{30}$

MA 11/16

Question 83 (JAM 2015 • Q13 • ID: JAM_2015_Q13) MCQ +2 M Neg: −0.67

Q.13 Let G be a nonabelian group. Let $\alpha \in G$ have order 4 and let $\beta \in G$ have order 3. Then the order of the element $\alpha\beta$ in G

(A) is 6

(B) is 12

(C) is of the form $12k$ for $k \geq 2$

(D) need not be finite

MA 7/16

Question 84 (JAM 2015 • Q11 • ID: JAM_2015_Q11) MCQ +2 M Neg: −0.67

Q.11 Let S_3 be the group of permutations of three distinct symbols. The direct sum $S_3 \oplus S_3$ has an element of order

(A) 4

(B) 6

(C) 9

(D) 18

Question 85 (JAM 2015 • Q1 • ID: JAM_2015_Q1) MCQ +1 M Neg: −0.33

Q.1 Suppose N is a normal subgroup of a group G. Which one of the following is true?

(A) If G is an infinite group then G/N is an infinite group
(B) If G is a nonabelian group then G/N is a nonabelian group
(C) If G is a cyclic group then G/N is an abelian group
(D) If G is an abelian group then G/N is a cyclic group

Question 86 (JAM 2014 • Q34 • ID: JAM_2014_Q34) MCQ +2 M Neg: −0.67

Q.34 Which one of the following conditions on a group G implies that G is abelian?

(A) The order of G is p^3 for some prime p
(B) Every proper subgroup of G is cyclic
(C) Every subgroup of G is normal in G
(D) The function $f: G \to G$, defined by $f(x) = x^{-1}$ for all $x \in G$, is a homomorphism

Question 87 (JAM 2014 • Q31 • ID: JAM_2014_Q31) MCQ +2 M Neg: −0.67

Q.31 Let S_n be the group of all permutations on the set $\{1, 2, \dots, n\}$ under the composition of mappings. For $n > 2$, if H is the smallest subgroup of S_n containing the transposition $(1, 2)$ and the cycle $(1, 2, \dots, n)$, then

(A) $H = S_n$
(B) H is abelian
(C) the index of H in S_n is 2
(D) H is cyclic

Question 88 (JAM 2014 • Q30 • ID: JAM_2014_Q30) MCQ +2 M Neg: −0.67

Q.30 Let G be a cyclic group of order 24. The total number of group isomorphisms of G onto itself is

(A) 7 (B) 8 (C) 17 (D) 24

Question 89 (JAM 2014 • Q7 • ID: JAM_2014_Q7) MCQ +1 M Neg: −0.33

Q.7 Let G be a group of order 17. The total number of non-isomorphic subgroups of G is

(A) 1 (B) 2 (C) 3 (D) 17

Question 90 (JAM 2013 • Q8 • ID: JAM_2013_Q8) MCQ +2 M Neg: −0.67

Let p be a prime number. Let G be the group of all 2×2 matrices over $\mathbb{Z}_p$ with determinant 1 under matrix multiplication. Then the order of G is

(A) $(p-1)p(p+1)$
(B) $p^2(p-1)$
(C) p^3
(D) $p^2(p-1)+p$

Question 91 (JAM 2012 • Q8 • ID: JAM_2012_Q8) MCQ +6 M Neg: −2.0

Q.8 Consider the quotient group $\mathbb{Q}/\mathbb{Z}$ of the additive group of rational numbers. The order of the element $\frac{2}{3}+\mathbb{Z}$ in $\mathbb{Q}/\mathbb{Z}$ is

(A) 2 (B) 3 (C) 5 (D) 6

Question 92 (JAM 2011 • Q10 • ID: JAM_2011_Q10) MCQ +6 M Neg: −2.0

Q.10 For $n\in\mathbb{N}$, let $n\mathbb{Z}=\{nk : k\in\mathbb{Z}\}$. Then the number of units of $\mathbb{Z}/11\mathbb{Z}$ and $\mathbb{Z}/12\mathbb{Z}$, respectively, are

(A) 11, 12 (B) 10, 11 (C) 10, 4 (D) 10, 8

MA- 2 / 40

Question 93 (JAM 2011 • Q9 • ID: JAM_2011_Q9) MCQ +6 M Neg: −2.0

Q.9 Let G denote the group of all 2×2 invertible matrices with entries from $\mathbb{R}$. Let

$$H_1 = \{A \in G : \det(A)=1\} \text{ and } H_2 = \{A \in G : A \text{ is upper triangular}\}.$$

Consider the following statements:

P : H_1 is a normal subgroup of G

Q : H_2 is a normal subgroup of G.

Then

(A) Both P and Q are true
(B) P is true and Q is false
(C) P is false and Q is true
(D) Both P and Q are false

Question 94 (JAM 2010 • Q12 • ID: JAM_2010_Q12) MCQ +6 M Neg: −2.0

Q.12 The number of 2×2 matrices over $\mathbf{Z}_3$ (the field with three elements) with determinant 1 is

(A) 24 (B) 60 (C) 20 (D) 30

Question 95 (JAM 2010 • Q3 • ID: JAM_2010_Q3) MCQ +6 M Neg: −2.0

Q.3 The number of elements of S_5 (the symmetric group on 5 letters) which are their own inverses equals

(A) 10 (B) 11 (C) 25 (D) 26

Question 96 (JAM 2009 • Q5 • ID: JAM_2009_Q5) MCQ +6 M Neg: −2.0

Q.5 The number of all subgroups of the group $(Z_{60}, +)$ of integers modulo 60 is

(A) 2 (B) 10 (C) 12 (D) 60

Question 97 (JAM 2008 • Q15 • ID: JAM_2008_Q15) MCQ +6 M Neg: −2.0

Q.15 Let G be a finite group and H be a normal subgroup of G of order 2. Then the order of the center of G is

(A) 0 (B) 1 (C) an even integer ≥ 2 (D) an odd integer ≥ 3

Question 98 (JAM 2008 • Q1 • ID: JAM_2008_Q1) MCQ +6 M Neg: −2.0

Q.1 The least positive integer n, such that $\begin{pmatrix} \cos\frac{\pi}{4} & \sin\frac{\pi}{4} \\ -\sin\frac{\pi}{4} & \cos\frac{\pi}{4} \end{pmatrix}^n$ is the identity matrix of order 2, is

(A) 4 (B) 8 (C) 12 (D) 16

Question 99 (JAM 2007 • Q6 • ID: JAM_2007_Q6) MCQ +6 M Neg: −2.0

6. Let G be an Abelian group of order 10. Let $S = \{g \in G : g^{-1} = g\}$. Then the number of non-identity elements in S is

(A) 5
(B) 2
(C) 1
(D) 0

Question 100 (JAM 2006 • Q13 • ID: JAM_2006_Q13) MCQ +6 M Neg: −2.0

13. Let G be a group of order 7 and $\phi(x) = x^4$, $x \in G$. Then ϕ is

(A) not one – one
(B) not onto
(C) not a homomorphism
(D) one – one, onto and a homomorphism

Question 101 (JAM 2005 • Q8 • ID: JAM_2005_Q8) **MCQ** **+6 M** **Neg: −2.0**

8. In the group $\{1, 2, \ldots, 16\}$ under the operation of multiplication modulo 17, the order of the element 3 is

(A) 4

(B) 8

(C) 12

(D) 16

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Groups (2005–2026 Archive) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+1	8 to 8
2	MCQ	+2	B
3	MCQ	+2	C
4	MCQ	+2	C
5	MCQ	+1	D
6	NAT	+2	0 to 0
7	NAT	+2	10 to 10
8	NAT	+2	3 to 3
9	MSQ	+2	B;D
10	MCQ	+2	A
11	MCQ	+1	A
12	MSQ	+2	B;D
13	MSQ	+2	B;C
14	MSQ	+2	A
15	MCQ	+2	B
16	MCQ	+2	A
17	MCQ	+1	A
18	MCQ	+1	A
19	MCQ	+1	D
20	MCQ	+1	A
21	NAT	+2	11 to 11
22	NAT	+1	6 to 6
23	NAT	+1	15 to 15
24	MSQ	+2	A, B, C
25	MCQ	+2	A
26	MCQ	+2	C
27	MCQ	+2	A
28	MCQ	+1	B
29	NAT	+2	6 to 6
30	NAT	+1	420 to 420
31	NAT	+1	8 to 8
32	MSQ	+2	A,C,D
33	MCQ	+2	B
34	MCQ	+2	A

Q. No.	Type	Marks	Official Key
35	MCQ	+1	B
36	NAT	+2	9
37	NAT	+1	4
38	NAT	+1	90
39	MSQ	+2	A, B
40	MCQ	+2	A
41	MCQ	+2	C
42	MCQ	+2	C
43	MCQ	+2	C
44	MCQ	+2	A
45	MCQ	+1	B
46	NAT	+2	3 to 3
47	NAT	+1	3 to 3
48	MSQ	+2	C;D
49	MCQ	+2	C
50	MCQ	+2	D
51	MCQ	+1	B
52	NAT	+2	4 to 4
53	NAT	+2	31 to 31 or 46 to 46
54	NAT	+1	99 to 99
55	MSQ	+2	B, C
56	MSQ	+2	A, C
57	MCQ	+2	D
58	NAT	+1	0 to 0
59	NAT	+1	4 to 4
60	MSQ	+2	B;C;D
61	MSQ	+2	A;D
62	MCQ	+2	D
63	MCQ	+2	C
64	MCQ	+2	A
65	MCQ	+1	B
66	NAT	+2	29.9 to 30.1
67	NAT	+1	7.9 to 8.1
68	NAT	+1	−0.01 to +0.01

Q. No.	Type	Marks	Official Key
69	NAT	+1	5.9 to 6.1
70	MSQ	+2	A, C
71	MSQ	+2	A, C
72	MCQ	+1	B
73	NAT	+2	3.0 to 3.0
74	NAT	+1	6.0 to 6.0
75	MSQ	+2	B;C
76	MCQ	+2	B
77	MCQ	+2	D
78	MCQ	+1	B
79	NAT	+2	84
80	NAT	+1	1
81	NAT	+1	3
82	MSQ	+2	A;D
83	MCQ	+2	D
84	MCQ	+2	B
85	MCQ	+1	C
86	MCQ	+2	D
87	MCQ	+2	A
88	MCQ	+2	B
89	MCQ	+1	B
90	MCQ	+2	A
91	MCQ	+6	B
92	MCQ	+6	C
93	MCQ	+6	B
94	MCQ	+6	A
95	MCQ	+6	D
96	MCQ	+6	C
97	MCQ	+6	C
98	MCQ	+6	B
99	MCQ	+6	C
100	MCQ	+6	D
101	MCQ	+6	D

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.